

Processing non-maximal readings of sentences with plural definites

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Abstract

Sentences with plural definites like ‘the circles’ are usually interpreted akin to universally quantified sentences (*maximal* reading). In some circumstances, however, they also allow for *non-maximal* readings, i.e imprecise readings that allow for exceptions. We report two mouse-tracking experiments that investigate the online derivation of these non-maximal interpretations of plural definites. Our results show that non-maximal readings are harder to process than maximal interpretations. This difficulty seems to be associated to hesitation in the decision-process rather than to a truly two-step derivation. Interestingly, we also find that experimental setup plays an important role in the availability and difficulty linked to these readings.

Keywords: definite plurals; non-maximal readings; mouse-tracking; processing; semantics

Introduction

Sentences that contain definite plural noun phrases such as ‘the students’ tend to be interpreted similar to sentences containing universal quantifiers like ‘all the students’:

- (1) I met the students. $\leadsto$ *I met all the students.*
- (2) a. The students came. $\leadsto$ *All the students came.*
b. The students didn’t come. $\leadsto$ *None of the students came.*

However, sentences with plural definites have specific properties that their universally quantified counterparts do not. To begin with, there is a range of situations that is not described properly by either the positive sentence nor its negated counterpart. For example, in a scenario where half of the students came, neither (2-a) nor (2-b) is a true description of the situation. Note that this situation could be described by corresponding universal negative sentence ‘All the students didn’t come’. The lack of complementary truth conditions for positive and negative sentences has been referred to as **homogeneity** (Löbner, 2000; Križ, 2016, a.o.)

Moreover, sentences with plural definites allow **non-maximal readings**, i.e readings that allow for exceptions. For instance, (2-a) could be accepted as true in a situation where a couple of the students did not come. The availability of these readings is sensitive to the context in which the sentence is uttered (Malamud, 2012; Križ, 2016; Križ & Spector, 2021).

Theoretical approaches

Recent semantic accounts of the meaning of sentences with plural definites have tried to draw a link between homogeneity and non-maximality (Križ, 2016; Križ & Spector, 2021; Bar-Lev, 2021). In particular, we are interested in two lines of analyses that we will rely on to draw hypotheses on the online processing of non-maximal readings.

One line of approaches to non-maximal readings proposes that the literal –semantic– meaning of a sentence with a plural definite is the maximal meaning, where the predicate must hold of all the members of the plurality. Here, the non-maximal readings arise as part of a pragmatic process whereby speakers accept the sentence as an appropriate description of a situation although it is not literally true (Lasnik, 1999; Križ, 2016). For example, Križ (2016) proposes to model sentences with plural definites as having trivalent truth-conditions. In other words, there are situations in which the sentence is neither true nor false, requiring a third truth value. Practically, this means that a sentence like (2-a) will be true in case all the students came, and false in case none of the students came, but will receive a third value in all the situations in between. In this system, a sentence that receives the third value can sometimes be accepted as *true enough* by speakers given some pragmatic factors. Specifically, if the non-maximal situation is as relevant for the purpose of the conversation as the situation in which the sentence is literally true.

Another line of analysis involves viewing sentences with plural definites as ambiguous with respect to multiple potential interpretations (Križ, 1996; Malamud, 2012; Križ & Spector, 2021). On the most simple version of this view, a sentence like (2-a) is ambiguous between a non-maximal reading, where at least some of the students came, and a maximal reading, where all of the students came. Since the maximal reading entails the non-maximal reading, both are true in a situation where all the students came, which makes the sentence clearly true for speakers in this case. However, in a situation where some, but not all of the students came, the sentence is true on one interpretation and false on the other. This results in an ambiguity resolution problem, where speakers have to decide on which interpretation of the sentence to choose.

Previous experimental work

Relatively little experimental work has been done to investigate homogeneity and non-maximality in sentences with plural definites. Most research has focused on the conditions under which non-maximal readings are accessible. For example, Križ and Chemla (2015) used a ternary truth judgment task (where participants have to decide whether a sentence is *clearly true*, *clearly false* or *neither true nor false*) to show that participants prefer to judge a sentence as *neither true nor false* in a case where the predicate of the sentence is true of some but not all of the members of the plurality.

In a more recent series of experiments, Romoli et al. (2024) have shown that the availability of non-maximal readings is actually sensitive to the context in which a sentence is used. Non-maximal readings become accessible (i.e., treated as true readings of the sentence) with the right contextual support.

As far as we know, the only study that has looked at *online* processing of these non-maximal readings is Schwarz (2013). In this study, participants were asked to judge whether a sentence of the form ‘the [SHAPES] are [COLOR]’ was a true or a false depiction of an array of colored shapes. When comparing ‘true’ responses in non-maximal cases (labeled ‘non-maximal’ responses) to ‘false’ responses for the same items (labeled ‘maximal’ responses), the author finds that the former take longer than the latter. Schwarz (2013) interprets this as showing that non-maximal readings are harder to derive than their maximal counterparts.

Current study

We aim at investigating the online derivation of non-maximal readings for sentences involving plural definites. Building on previous results from Schwarz (2013), we expect that deriving a non-maximal reading for a sentence with a definite plural is more costly than deriving a maximal reading of the same sentence. Besides replicating the pattern of cognitive cost associated with non-maximal interpretations, we seek to shed light on the underlying processes that lead to this cost—a question that has remained unexplored in previous research.

While the literature on non-maximality is agnostic about the *online* derivation of non-maximal interpretations, we propose two competing hypotheses about the processing of these readings, each compatible with one of the aforementioned theoretical accounts of non-maximality. These hypotheses are built on the idea that there is a parallelism between how these readings are generated and how they are actually obtained during parsing.

Under one view, non-maximal interpretations are harder to process because they involve a two-step derivation. Unlike maximal interpretations—which are thought to be the default interpretation of sentences with plural definites—, non-maximal readings are obtained from maximal readings via pragmatic reasoning (Laserson, 1999; Križ, 2016). In this view, comprehenders always start by computing a maximal reading of the sentence, and only as a second step the non-maximal reading is derived via pragmatic reasoning. For a

sentence like (2-b), in a context where only some of the students came, one would first access the strong reading, where all the students came and decide that the sentence is false—or at least not true, before admitting a pragmatic, *true enough* reading, where some of the students didn’t come, resulting in a change of decision.

A similar processing hypothesis has been proposed for scalar implicatures (Tomlinson Jr, Bailey, & Bott, 2013; Huang & Snedeker, 2018,a.o) where a sentence like ‘Some elephants are mammals’ has been claimed to be first interpreted as ‘Some and maybe all elephants are mammals’ and only later strengthened into ‘Some but not all elephants are mammals’.

Another possible hypothesis is that maximal and non-maximal readings are derived independently from each other. That is, sentences with plural definites are truly ambiguous between two independent readings. When comprehenders encounter these sentences, both readings compete in parallel, leading to general hesitation. Differences in cognitive cost are expected because non-maximal readings are less likely, or dispreferred, compared to their maximal counterparts.

In order to tease these hypotheses apart, we run two different experiments that use a mouse-tracking methodology to investigate the processing of non-maximal interpretations.

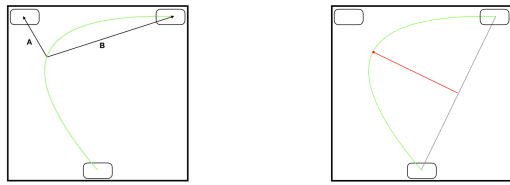
Mouse tracking as a window into non-maximal readings

Mouse tracking has become a widely used method in cognitive sciences to study the dynamics of decision-making (see summary in Freeman, Dale, & Farmer, 2011). This research is based on the key assumption that hand movements are planned and executed alongside decisions (Wojnowicz, Ferguson, Dale, & Spivey, 2009). As a result, mouse trajectories are thought to provide insight into cognitive processes in a way that response times alone cannot.

Specifically, when a participant selects a response by clicking a button, the time taken to click can serve as a rough indicator of processing cost. However, the trajectory of the mouse movement leading to that button may reveal additional information, such as whether participants were confident from the start, hesitated, or underwent a radical shift in decision (Maldonado, Dunbar, & Chemla, 2019).

In this study, we use mouse tracking to investigate the online derivation of non-maximal readings in sentences containing plural definites. This approach has previously been used to explore similar linguistic phenomena, such as the derivation of scalar implicatures or presuppositional content (Tomlinson Jr et al., 2013; Schneider, Schonard, Franke, Jäger, & Janczyk, 2019).

We will focus on two specific parameters of mouse trajectories that can help us distinguishing two-step decision processes from mere uncertainty: (i) the maximal deviation from the ideal (straight-line) trajectory and (ii) the maximal log ratio of the distance to the target versus the distance to the competitor. An illustration of these measures is provided in Fig. 1.



(a) Maximal Log-Ratio (b) Maximal Deviation

Figure 1: Mouse tracking measures

These measures were chosen because they are strong predictors of whether a trajectory is diverted toward the competitor (Maldonado et al., 2019).

Experiment 1

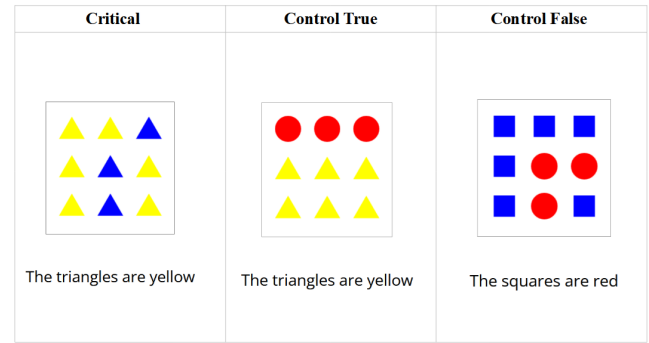
Method

Design and Materials Participants are presented with sentence-picture pairs and are instructed to decide whether the sentence is a true or false description of the picture. During this task, their mouse-trajectories are recorded (see Procedure below). Each picture displays an array of nine colored shapes (squares, circles, or triangles in red, blue, or yellow). The sentences, which contain plural definites, have form ‘The [SHAPES] are [COLOR]’.

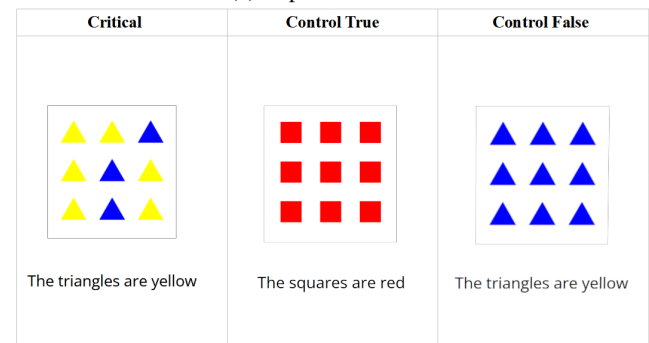
There are three types of trials (see Fig. 2a). In Critical trials, the picture displays an array of nine tokens of the shape mentioned in the sentence, where six of them are of the color used in the sentence, and three are a different color. This picture makes the sentence true under a non-maximal interpretation, and false under any other reading. We also include two types of control trials. In Control True trials, the picture makes the sentence true under a maximal interpretation; in Control False trials the picture makes the sentence false under all its readings. Note that all pictures involve two different colors, ensuring that critical trials are not visually more complex than controls. This is an important distinction from Schwarz (2013).

Each participant sees three items of each condition. All pictures are automatically generated and the order of the shapes in the array is randomized. The experiment was pre-registered in <https://tinyurl.com/mse9ev2m>.

Procedure The experiment is run in a browser. The procedure is exemplified in Fig. 3. Each trial begins with a screen displaying the picture. Participants have to click on the button that says [SHOW THE SENTENCE] to make the sentence visible and be able to respond by clicking on the TRUE/FALSE buttons. Participants have 6 seconds to select their response. We set this time limit to put participants under pressure and ensure they would begin moving their mouse as soon as the sentence appears. To further enforce this time pressure, we added a timer sound that activates 3 seconds into the trial and progression bars on the sides of the screen that in-



(a) Experiment 1



(b) Experiment 2

Figure 2: Illustration of Critical and Control Trials.

dicating how much time there is left to answer. The experiment was coded using the jsPsych (de Leeuw, Gilbert, & Luchterhandt, 2023) and we collected responses, response times and mouse trajectories (as triples of x coordinate, y coordinate and time) for each trial.

Participants We recruited 275 (non color-blind) English-speaking participants through Prolific. Participants were compensated at a rate of 10 £/h for taking the experiment (about 0.84 £ for a 5 minutes participation). Per our preregistration, we included only those who reported using a computer mouse and being native speakers of English at the end of the experiment. We also excluded participants who did not correctly answer at least one of the controls. We analyzed a total of 161 participants.

Results

Recall that we are interested in the derivation of non-maximal readings for Critical trials, compared to the derivation of maximal readings for Control True trials. Consequently, we only analyze the response times and mouse trajectories of Critical and Control True trials in which participants answered ‘True’. The overall acceptance rate was 73% (se = 3.2pp) in Critical trials. 32 participants were excluded from the analysis because of not having any ‘True’ response in Critical trials.

Response times Log-transformed response times for True Critical and Control trials are shown in Fig 7 (left side). A linear mixed model with Trial (Critical vs. Control True, treat-

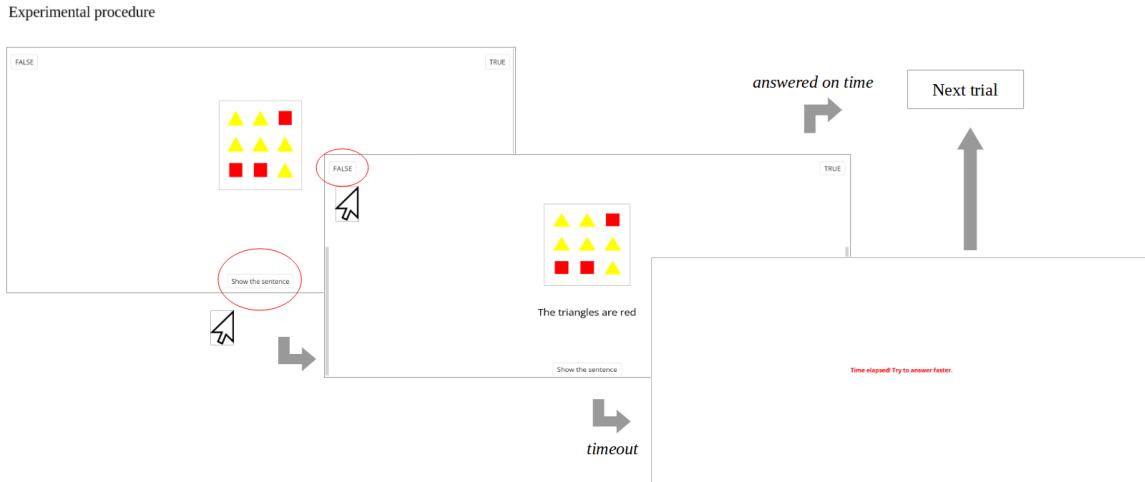


Figure 3: Experimental procedure

ment coded with Control as baseline) as the predictor and a random intercept per participant showed no statistically significant effect of Trial on response times, failing to confirm our hypothesis that non-maximal readings are more cognitively costly than maximal readings, and thus failing to replicate Schwarz (2013)’s findings.

Mouse trajectories Mouse-trajectories were pre-processed and normalized on time (on 101 time steps) using the mouse-trap R package (Wulff, Kieslich, Henninger, Haslbeck, & Schulte-Mecklenbeck, 2021).

In order to assess the degree of trajectory deviation, we compute two measures: the maximal deviation and the maximal log ratio of distance to target and to competitor. For each measure, we computed, for each participant, the mean difference between Critical and Control trials and then calculated the overall mean difference. To assess whether the observed difference was likely due to chance, we used a permutation test. Specifically, we generated 1000 random permutations of the data by shuffling the condition labels within each participant. We computed the overall mean difference for each permutation and then used a one-sample t-test to assess how likely it is that the real mean comes from the permuted distribution. The permutation test did not reveal any significant difference between the critical and control trials, so we couldn’t confirm our hypothesis that non-maximal readings are derived via a two-step process.

Discussion

Experiment 1 yielded a surprisingly high acceptance rate of non-maximal interpretations in Critical trials, compared to the results of (Schwarz, 2013; where the acceptance rate was around 30%).

One important difference between our experiment and Schwarz (2013)’s was the nature of the controls used in the

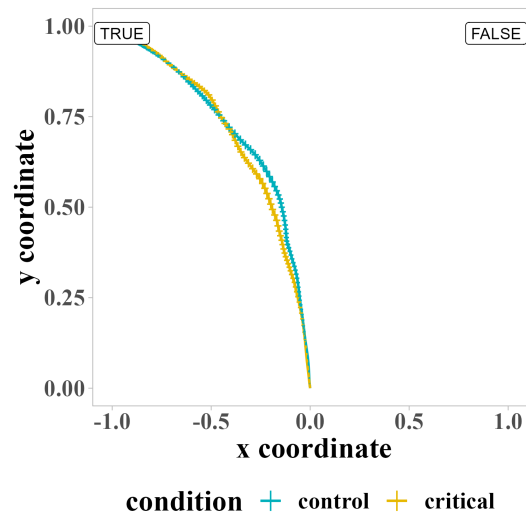


Figure 4: Mean trajectories for critical and control conditions in experiment 1. Error bars represent standard error over x and y coordinates.

experiment. In his experiment, control trials involved pictures containing an array of nine equal shapes, all of them of the mentioned color in the sentence or none of them of the mentioned color. In our case, we chose to make the controls more difficult by making the picture show two different types of shapes.

Our choice of controls might have affected the results in two ways. First, the control items may have ended up being as complex as Critical trials precisely *because* they contained two shapes instead of one. This added complexity might have overshadowed any additional difficulty arising from non-maximal readings, leading to no overall observed

difference.

Second, we know that the context of utterance of sentences with plural definites affect the availability of non-maximal readings (Malamud, 2012; Križ, 2016; Romoli et al., 2024). It's possible then that in the presence of controls involving two colors might have provided a context where non-maximal readings of the sentence (where are two colors for the same shape) are more pertinent, salient, or just easier to access.

In sum, with the design of Experiment 1, we were not able to replicate the findings from Schwarz (2013) that non-maximal readings are more cognitively costly than maximal readings. As a result, we were unable to adequately assess the underlying processing of these sentences. In Experiment 2, we return to the original materials used in Schwarz (2013).

Experiment 2

Method

Experiment 2 is a partial replication of Experiment 1, aimed at controlling the potential influence of control pictures in our findings. We speculate that controls in Experiment 1 may have made non-maximal readings more accessible relative to true controls, thereby masking the potential difficulty of non-maximal readings. Accordingly, in Experiment 2, we changed the pictures in control items so they feature an array of nine objects of the same shape and color (see Fig. 2b). The design and procedure remained otherwise identical to those in Experiment 1.

In addition to testing the same hypotheses as in Experiment 1, Experiment 2 is designed to assess the effect of control pictures in the access to non-maximal readings. We expect that this change in controls will result in a significant decrease of non-maximal readings.

This experiment was pre-registered in <https://tinyurl.com/ycyxeshd>.

Participants

We recruited a total of 292 participants through Prolific, using the same filters and exclusion criteria as in experiment 1. We kept 201 participants for the analysis.

Results

Acceptance rate for the Critical condition in Experiment 2 was 31% (SE = 3.5pp). 130 participants were excluded from the analysis because they didn't have any 'True' response in Critical trials.

Response times Fig. 7 (right side) displays the log-transformed response times for Experiment 2 for 'true' responses in Critical and Control trials. A linear mixed-effects model, similar to the one applied on Experiment 1 data, reveals a significant effect of Trial ($\beta = 0.25$, $p > 0.05$), with longer reaction times in the Critical compared to Control trials.

Mouse trajectories Average mouse trajectories for 'true' responses in both conditions are shown in Fig. 6. To evaluate

the hypothesis that non-maximal readings involve a two-step derivation, we once again compared Critical and Control trajectories with respect to their maximal deviation with respect to the ideal path and their maximal log-ratio from competitor to target. The results for the permutation tests are provided in Fig. 5, where it can be observed that the difference between the two types of trials is not statistically significant for any of the two measures.

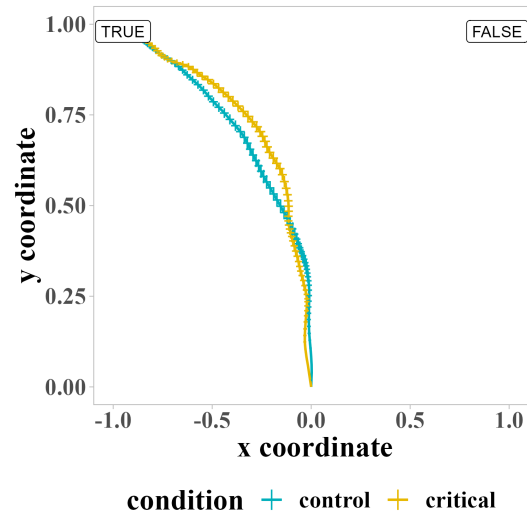


Figure 6: Mean trajectories for critical and control conditions in experiment 2. Error bars represent standard error.

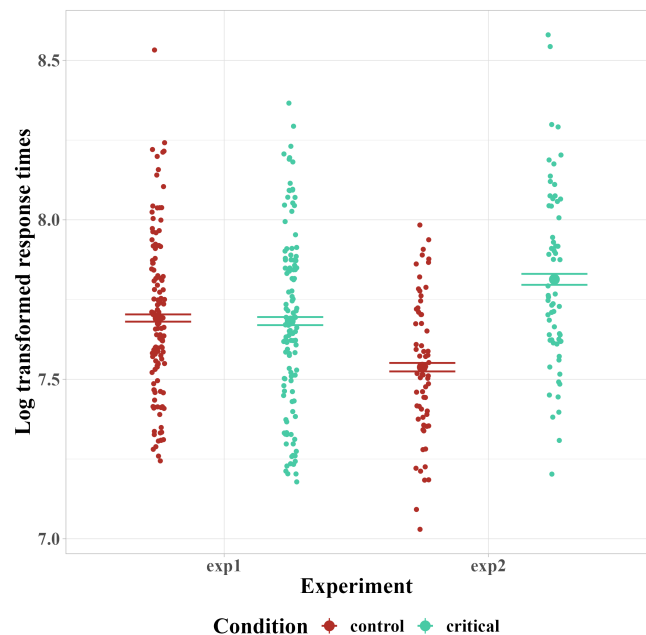


Figure 7: Response times for both experiments. Error bars represent standard error and small dots represent participant's means.

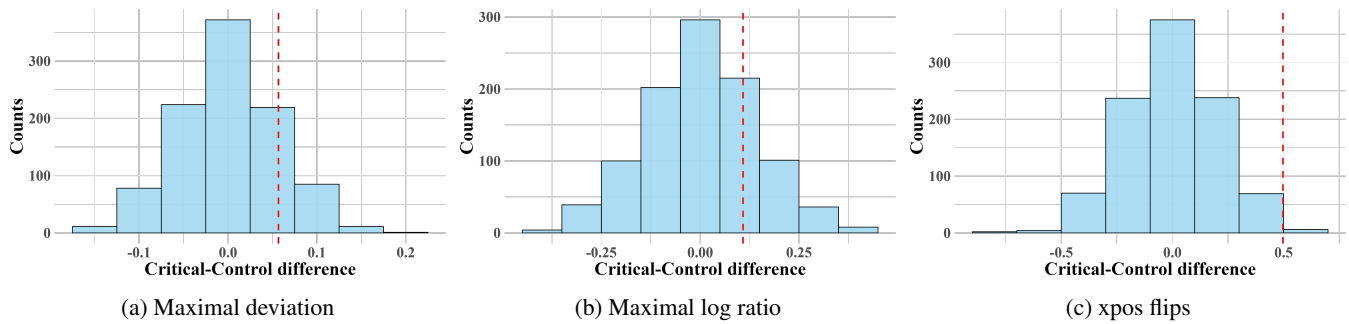


Figure 5: Permutation tests in Experiment 2. Red dotted line represents the actual mean difference between critical and control conditions.

Discussion

Experiment 2 aimed to determine whether the lack of evidence for a difference between maximal and non-maximal readings in Experiment 1 was due to the properties of the control trials. Specifically, we examined whether these controls played a role in the absence of evidence for the difficulty of non-maximal interpretations.

The findings in Experiment 2 support this idea. First, the acceptance rate of non-maximal readings in Experiment 2 is considerably lower than in Experiment 1, suggesting that the controls used in the experiment did have an effect on the general availability of non-maximal readings. Unsurprisingly, when participants do access these readings, it takes them more time. Indeed, the response times results in Experiment 2 replicate Schwarz (2013), indicating that non-maximal readings of sentences with plural definites are more costly to derive than maximal readings.

The replication of previous results allows for a more careful interpretation of mouse-tracking trajectories. The two measures we selected to test the existence of an underlying two-step process did not show sensitivity to the difference in difficulty between maximal and non-maximal interpretations.

Since we didn't find evidence for a two-step process but found evidence for longer processing times, we decided to further explore the data by analyzing mouse-tracking measures that were not part of our original hypotheses. We specifically chose to look at the average number of x flips (i.e. changes in direction along the x-axis). Unlike maximal deviation and maximal log-ratio, which are measures of curvature, xpos flips are a measure of the complexity of trajectory (Wulff et al., 2021), which could indicate a general hesitation in the decision-making process. When performing the permutation test on this measure, we found a statistical difference between Critical and Control trials (see Fig. 5c). However, this result is exploratory and should be confirmed by further experiments.

Let us finally mention that while our findings indicate that the controls influence the general availability and ease of non-maximal readings, the underlying reason why this happens remains an open question.

Conclusion

This study aimed at providing more insights into the processing of non-maximal readings of plural definites. Our first goals were twofold: confirming that deriving non-maximal readings is indeed more cognitively costly than deriving maximal readings, and assessing the underlying cognitive process that gives rise to this extra difficulty.

Our results replicate the findings from Schwarz (2013) that non-maximal readings indeed involve a higher processing cost. However, our mouse-tracking results fail to show evidence for a two-step derivation in the processing of non-maximal readings. Instead, we found evidence for a general hesitation. These findings are more compatible with a view in which sentences with plural definites give rise to an ambiguity between distinctive maximal and non-maximal readings, than with a view in which a literal maximal reading is derived first and the non-maximal reading is derived as a second step. Further research is needed to confirm whether this is the case. Our results thus contribute a first step in better understanding the underlying mechanisms involved in the online derivation of non-maximal readings.

Interestingly, our study also revealed that the experimental setup, in particular the alternative situations participants encounter in the experiment, has an interesting impact on the availability of non-maximal readings. This impact should be considered in further research on this phenomenon.

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